KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE

KUMASI, GHANA

PROJECT PROPOSAL

HarvestLink:

A Mobile Agritech Logistics Application for Connecting Smallholder Farmers to Markets in Ghana

Course Code: CSM 366

Computer Science Mini Project

Supervisor

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Submitted By

Team Member	Index Number	Role
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Submission Date

August 10, 2026

1. Executive Summary

Ghana loses a large share of its perishable harvest before it ever reaches a buyer. In the Bono and Ashanti regions, farmers watch tomatoes, maize, and cassava spoil at the farm gate, not because nobody wants to buy them, but because nothing connects a farmer with produce ready today to a buyer looking for it today. Transport is informal, pricing is a guessing game, and large stretches of these regions sit in mobile network dead zones for parts of the day. HarvestLink is our attempt to close that gap. It's a React Native app that lets a farmer log a harvest with no signal at all, using WatermelonDB to hold the data on the device until it can sync to our PostgreSQL backend through Prisma. Once synced, a rule-based matching engine pairs that harvest with nearby buyers by crop type, quantity, and distance, and buyers track the delivery in real time once a match is confirmed. By submission, we plan to have a working Android build, a documented backend, and a tested matching workflow ready to demo on August 10, 2026.

2. Problem Statement

Ghana's agricultural supply chain is fragmented, and smallholder farmers are the ones who absorb the cost of that fragmentation: little market information, patchy logistics, and almost no digital tools built for how they actually work. Two problems stood out in our research.

2.1 Market Access and Logistics

Most smallholder farmers sell through intermediaries and often end up with only 30 to 40 percent of what the consumer eventually pays, largely because they have no real-time view of market prices (FAO, 2021). Getting produce to market isn't much better: transport is arranged informally, with no visibility into whether a vehicle is even available, so produce sits waiting and often arrives in worse condition than it left in.

2.2 Post-Harvest Loss and the Digital Gap

Ghana loses an estimated 20 to 40 percent of perishable produce between harvest and sale, mostly down to poor storage, rough transport conditions, and logistics delays (World Bank, 2020). Mobile phones are everywhere, over 60 percent penetration (GSMA, 2023),

but the tools that exist are built for large agribusinesses, assume connectivity that rural areas don't have, or simply weren't designed with the Ghanaian context in mind.

3. Project Objectives

1. Build a working Android app in React Native that covers farmer onboarding, produce listing, buyer matching, and delivery tracking, ready by the end of the semester.
2. Cut simulated post-harvest loss by at least 30 percent in a controlled test, using buyer-matching and timely listings to get produce moving faster.
3. Hit a System Usability Scale (SUS) score of 70 or above, tested with at least 10 real participants, not just the team.
4. Make at least five core workflows work offline using WatermelonDB on the device, syncing to PostgreSQL once a connection comes back.
5. Write up a Software Requirements Specification, a UI prototype, and a test report, all to a standard we'd be comfortable submitting.
6. Demo the working product to the supervisor and peer evaluators on the scheduled presentation date.

4. Proposed Solution

HarvestLink is a React Native app for Android, talking to a Node.js and Express backend with PostgreSQL as the source of truth, through Prisma. On the device, WatermelonDB holds data locally so a farmer can log a harvest with no signal at all; it syncs to the server the moment a connection comes back. The app is built around five modules.

Module	Description
Onboarding	Farmer and buyer sign-up with phone number authentication via MTN MoMo or Vodafone Cash. Role-based access control.
Marketplace	Searchable, filterable produce listings with quantity, pricing, and availability dates. Buyers browse and express interest.
Buyer Matching	Rule-based engine pairing farmers with buyers by produce type, quantity,

	proximity, and transaction history.
Delivery Tracking	Real-time GPS tracking with live driver location, ETA, and delivery status for both farmer and buyer.
Profile Management	Dashboard for delivery history, earnings, and ratings. Reads and writes locally via WatermelonDB when offline.

5. Scope of Work

In Scope	Out of Scope
Figma prototype, all 8 screens	iOS build
Android app: React Native, NativeWind, Zustand, and WatermelonDB	Live payments or mobile money payout
Node.js/Express API with OTP login	ML-based demand forecasting
PostgreSQL schema via Prisma	Production deployment / Play Store listing
Rule-based buyer-matching	Fleet or vehicle management
Google Maps SDK for delivery tracking	—
Offline sync: device to Postgres on reconnect	—
Unit tests, UAT, and SRS write-up	—

6. Methodology

The team will work in two-week sprints, Agile-style, so supervisor feedback can be folded in early rather than discovered at the end. The stack below is what we are building with, and the reasoning behind each choice.

Layer	Technology	Why we picked it
UI/UX Design	**Figma**	Standard for prototyping and handing off designs to developers
Mobile App	**React Native**	One codebase for Android, and the team already

		knows it
Styling	**NativeWind**	Tailwind-style classes inside React Native, faster to style screens
State Management	**Zustand**	Does what Redux does with a fraction of the setup
Offline Storage	**WatermelonDB**	Lets farmers log a harvest with no signal; syncs to Postgres later
Backend / API	**Node.js + Express**	Familiar, lightweight, same language as the frontend
Database	**PostgreSQL**	Farmer and buyer records are relational; this is the right fit
ORM	**Prisma**	Type-safe queries and migrations, less time spent writing SQL by hand
Mapping	**Google Maps SDK**	GPS tracking and routes for delivery tracking
Build Tooling	**Metro + Gradle**	Metro bundles the JS, Gradle builds the Android binary
Version Control	**Git + GitHub**	Branching and pull requests for the three of us to work in parallel
Project Management	**Jira**	Sprint boards and a backlog we can point to in the report
Testing	**Jest + manual UAT**	Unit coverage, plus real people testing real screens

7. Timeline and Work Plan

The semester runs from Saturday, May 23, 2026 to Saturday, September 5, 2026, with exams starting Monday, August 17, 2026. We've planned five phases across eleven weeks, aiming to submit by August 10, 2026, a week clear of exams.

Phase	Weeks	Key Activities	Deliverable

Phase 1: Planning	Weeks 1–2 (May 23 – Jun 5)	Project scoping; literature review; requirements gathering; supervisor consultation; SRS drafting.	SRS v1.0
Phase 2: Design	Weeks 3–4 (Jun 6 – Jun 19)	Figma wireframes; high-fidelity prototype (all 8 screens); design system documentation; supervisor review.	Figma Prototype
Phase 3: Development	Weeks 5–8 (Jun 20 – Jul 17)	Sprint 1 (Onboarding + Auth); Sprint 2 (Marketplace); Sprint 3 (Buyer Matching); Sprint 4 (Delivery Tracking + WatermelonDB Offline Sync).	Functional App
Phase 4: Testing	Weeks 9–10 (Jul 18 – Jul 31)	Unit testing; integration testing; UAT sessions with 10+ participants; SUS evaluation; bug fixing.	Test Report
Phase 5: Submission	Week 11 (Aug 1 – Aug 10)	Final documentation; project report writing; demonstration preparation; submission at least one week before exams begin.	Final Report + Demo

8. Team Members and Responsibilities

Name	Role	What they're actually doing
Sanker Mwinwieme Protus	Backend	Backend architecture, PostgreSQL schema and Prisma migrations, the buyer-matching logic, API design, and pulling the final report together.
Owusu Annor Benedict Adu Acheampong	Team Lead / UI/UX Designer	Figma design system, all screen prototypes, user research, running usability tests, and accessibility review.
Owusu My-Frank	Frontend Developer	React Native screens with NativeWind,

		Zustand state, Google Maps integration, and the WatermelonDB offline layer.

9. Expected Outcomes and Impact

9.1 Measurable Deliverables

1. A high-fidelity Figma prototype encompassing eight application screens, designed against WCAG 2.2 AA accessibility guidance.
2. A functional Android mobile application implementing all five core modules: Onboarding, Marketplace, Buyer Matching, Delivery Tracking, and Profile Management.
3. A documented test report demonstrating a System Usability Scale score of 70 or above, with evidence from a minimum of 10 test participants.
4. A complete Software Requirements Specification and project report submitted to the Department of Computer Science, KNUST.

9.2 Anticipated Societal Impact

1. Increased farmer income through direct buyer-to-farmer price negotiation, reducing dependency on intermediaries.
2. Reduced post-harvest losses through timely buyer matching and coordinated logistics, potentially saving up to 30 percent of perishable produce value.
3. Improved market transparency through real-time produce listings and price visibility accessible to both farmers and buyers.

10. Conclusion

HarvestLink takes a real problem in Ghana's agricultural sector and applies mobile software engineering to it directly: poor market access, fragmented logistics, and a lack of digital tools built for how smallholder farmers actually operate.

We're confident we can deliver a working prototype that meets the requirements of CSM 366, and one that could genuinely stand on its own outside the classroom. We have the

skills across the team to cover mobile development, UI/UX, backend and database work, and testing, and we're asking the supervisor's approval to start building.

11. References

Food and Agriculture Organisation of the United Nations (FAO). (2021). The state of food and agriculture 2021: Making agrifood systems more resilient to shocks and stresses. FAO. https://doi.org/10.4060/cb4476en

GSMA Intelligence. (2023). The mobile economy: Sub-Saharan Africa 2023. GSMA. https://www.gsma.com/mobileeconomy/sub-saharan-africa/

Ministry of Food and Agriculture, Ghana (MoFA). (2022). Agriculture in Ghana: Facts and figures 2021. Government of Ghana.

World Bank. (2020). Reducing post-harvest losses in Ghana: A review of evidence and policy options. World Bank Group.

W3C Web Accessibility Initiative. (2023). Web Content Accessibility Guidelines (WCAG) 2.2. https://www.w3.org/TR/WCAG22/